

PALKIN

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Education

Bachelor of Technology in Industrial Internet of Things

Aug 2023 – May 2027

Vivekananda Institute of Professional Studies, New Delhi **CGPA: 8.75**

Delhi, India

D.A.V Public School, Rohtak, Haryana

Class 10th: **93%** Class 12th: **90%**

Haryana, India

Technical Skills

- **Languages:** Python, SQL
- **Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn
- **Frameworks:** PyTorch, TensorFlow, Flask
- **Tools:** Git, GitHub, Jupyter Notebook

Projects

- **Jet Engine Predictive Maintenance** *Python, Scikit-learn, XGBoost, Streamlit*
 - Built end-to-end ML pipeline on **NASA CMAPSS dataset** (20,631 readings, 21 IoT sensors) to predict Remaining Useful Life (RUL) of jet engines.
 - Engineered RUL target variable; removed low-variance sensors via std deviation analysis and applied MinMax normalization.
 - Compared 3 models (Linear Regression, Random Forest, XGBoost) — Random Forest achieved best **$R^2 = 0.63$** , **MAE = 29.52 cycles**.
 - Deployed interactive **Streamlit** web app with real-time RUL prediction and color-coded health alerts.
- **Autonomous Q-Learning Agent (Reinforcement Learning)** *Python, Gymnasium, NumPy, Matplotlib*
 - Built a **Q-Learning agent** on Taxi-v3 environment that learns to navigate, pick up, and drop off passengers with **zero prior knowledge**.
 - Implemented **Bellman Equation** with epsilon-greedy strategy achieving **85% success rate** after 2000 training episodes.
 - Reward improved from **-420 to +7.32** over training, demonstrating convergence to optimal policy.
 - Visualized training performance using **Matplotlib** and saved Q-table for reproducible evaluation.
- **Personal Portfolio Website** *Python, Flask, HTML, CSS, JavaScript*
 - Built a **full-stack portfolio website** using Python (Flask) as backend to showcase projects and skills.
 - Designed **responsive UI** with HTML/CSS featuring smooth scrolling, hover effects, and project cards.
 - Integrated a **contact form** with backend email handling using Python's smtplib library.

Certifications

- Supervised Machine Learning: Regression and Classification **DeepLearning.AI Stanford Online** (Coursera) — Apr 2026.
- Advanced Learning Algorithms **DeepLearning.AI Stanford Online** (Coursera) — Apr 2026.
- Unsupervised Learning, Recommenders, Reinforcement Learning **DeepLearning.AI Stanford Online** (Coursera) — Apr 2026.
- Introduction to Internet of Things (IoT) — **Elite Certification** — **Consolidated Score: 83 percentile** NPTEL (IIT Kharagpur) — Jan–Apr 2025.